TT-P-86H
July 19, 1987
SUPERSEDING
TT-P-86G
March 19, 1974

FEDERAL SPECIFICATION

PAINT, RED-LEAD-BASE, READY-MIXED

This specification was approved by the Commissioner, Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers red-lead-base, ready-mixed paint, for use as a primer on iron and steel surfaces. The two classes are representative of materials to be used in specified areas, when air quality regulations are applicable (see 6.6).

1.2 Classification. The paint shall be of the following types and classes (see 6.2 and 6.5):

Type I - Red lead-linseed oil paint
Type II - Red lead mixed pigment-alkyd varnish-linseed oil paint
Type III - Red lead-alkyd varnish paint
Type IV - Red lead mixed pigment-phenolic varnish paint
Class 1 - Volatile organic compounds (VOC) limited
Class 2 - Not photochemically reactive

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications

TT-T-291 - Thinner, Paint, Mineral Spirits, Regular and Odorless
PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of

Federal Standards

FED-STD-141 - Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing
FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of

AMSC N/A FSC 8010

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Federal Regulations:

Code of Federal Regulations (CFR)

Title 29 - Labor
Parts 1900-1910 - Occupational Safety and Health Adminstration (OSHA), Department of Labor

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM):

D50 - Chemical Analysis of Yellow, Orange, Red, and Brown Pigments Containing Iron and Manganese, Methods of
D71 - Relative Density of Solid Pitch and Asphalt (Displacement Method), Test Method for
D562 - Consistency of Paints Using the Stormer Viscometer, Test Method for
D563 - Phthalic Anhydride Content of Alkyd Resins and Resin Solutions, Test Method for
D600 - Liquid Paint Driers, Specification for

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

D1208 - Common Properties of Certain Pigments, Test Methods for
D1210 - Fineness of Dispersion of Pigment-Vehicle Systems, Test Method for
D1398 - Fatty Acid Content of Alkyd Resins and Alkyd Resin Solutions, Test Method for
D1475 - Density of Paint, Varnish, Lacquer, and Related Products, Test Method for
D1542 - Qualitative Tests for Rosin in Varnishes, Method of
D1544 - Color of Transparent Liquids (Gardner Color Scale), Test Method for
D1545 - Viscosity of Transparent Liquids by Bubble Time Method, Test Method for
D1642 - Elasticity or Toughness of Varnishes, Test Methods for
D1643 - Gas Checking and Draft Test of Varnish Films, Test Methods for
D1644 - Nonvolatile Content of Varnishes, Test Methods for
D1959 - Iodine Value of Drying Oils and Fatty Acids, Test Method for
D2090 - Clarity and Cleanness of Paint Liquids, Test Method for
D2245 - Identification of Oils and Oil Acids in Solvent-Reducible Paints, Method for
D2369 - Volatile Content of Coatings, Test Method for
D2697 - Volume Nonvolatile Matter in Clear or Pigmented Coatings, Test Method for
D2698 - Determination of the Pigment Content of Solvent-Reducible Paints by High-Speed Centrifuging, Method for
D3924 - Standard Environment for Conditioning and Testing Paint, Varnish, Lacquer, and Related Materials, Specification for
D3925 - Sampling Liquid Paints and Related Pigmented Coatings, Practice for
D4017 - Water in Paints and Paint Materials by Karl Fischer Method, Test Method for
D4287 - Determination of Viscosity of Paints and Varnishes at a High Rate of Shear by the ICI Cone/Plate Viscometer, Test Method for
E28 - Softening Point by Ring-and-Ball Apparatus, Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Illinois Administrative (Ill. Adm.) Code:

Title 35 - Environmental Protection
Section 211.122 - Definitions
Section 215.561 - Architectural Coatings

(Application for copies should be addressed to the Illinois Environmental Protection Agency, 2200 Churchill Road, Springfield, IL 62794-9276.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

South Coast Air Quality Management District (SCAQMD) Rules and Regulations:

Rule 1107 - Coating of Metal Parts and Products
Rule 1113 - Architectural Coatings

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 E. Flair Drive, El Monte, CA 91731-2884.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract of purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2).

3.2 Color. Types I, III, and IV paint shall have the characteristic red lead color, and type II paint shall have the color characteristic of a blend of red lead and red iron oxide.

3.3 Vehicle.

3.3.1 Vehicle, types I and II. The vehicle shall be linseed oil (type I), or alkyd-varnish-linseed oil (type II).

3.3.2 Phenolic varnish, type IV. The resin and varnish for type IV paint (see 6.3.2) shall conform to the requirements of table I, when tested as specified.

3.3.2.1 Hot water resistance. The phenolic varnish film after recovery from immersion in boiling water shall show no evidence of blistering, dulling, whitening, or loss of adhesion.

3.3.2.2 Alkali resistance. The phenolic varnish film after recovery from alkali immersion shall show no change in appearance such as haze spots or pits, or loss of adhesion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

TABLE I. Phenolic varnish, type IV.

Characteristics	Requirements			Test Methods	
	Min	Max	Para	ASTM	Para
Resin					
Specific gravity	1.03	1.06	-	D71	-
Softening point, °C (°F)	85(185)	99(210)	-	E28, 7.1.3	-
Color number (50 percent toluene solution)	-	10	-	D1544	-
Ash, percent	-	0.03	-	D1208, Sec 4	-
Varnish					
Nonvolatile content, mass percent	59	61	-	D1644, Meth A	-
Viscosity, stokes	140	200	-	D1545	-
Color number	-	12	-	D1544	-
Density, kg/L (lb/gal)	0.90 (7.5)	-	-	D1475	-
Kauri reduction, percent	120	-	-	D1642, Meth B	4.4.1.1, 4.4.1.2
Hot water resistance, hours	7	-	3.3.2.1	-	4.4.1.1, 4.4.1.3
Alkali resistance, hours	7	-	3.3.2.2	-	4.4.1.1, 4.4.1.4
Gas checking	Uniform after test		-	D1643, Meth A	4.4.1.1
Appearance	Clear, free from sediment		-	D2090	4.4.1.1

3.3.3 Volatile vehicle.

3.3.3.1 Class 1. The VOC content shall be limited as defined by SCAQMD Rule 1107 or 1113, for the intended use.

3.3.3.2 Class 2. The composition of the volatile vehicle shall allow the coating to be not photochemically reactive as defined by 35 Ill. Adm. Code 211.122 and 215.561, for the intended use.

3.4 Paint. The paint shall conform to the requirements of table II, when tested as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

TABLE II. Paint.

Characteristics	Requirements									Test Methods		
	I		II		III		IV					
	Min	Max	Min	Max	Min	Max	Min	Max	Para	ASTM	FTMS 141	Para
Analysis												
Pigment, mass percent 1/	77.0	-	66.0	-	67.0	-	65.0	-	3.5	D2698	-	-
Red lead, Pb_3O_4, mass percent 2/	96.5	-	62.5	-	96.5	-	82.0	-	-	-	7071, 7073	4.4.2.1
Iron oxide, Fe_2O_3, mass percent 2/	-	-	12.5	-	-	-	-	-	-	D50	-	-
Siliceous extender and mica, mass percent 2/	-	-	-	22.0	-	-	-	15.5	-	-	-	4.4.2.2
Vehicle												
Nonvolatile vehicle, mass percent 3/	65.0	-	56.0	-	40.0	-	44.0	-	-	D2698	-	-
Identification, types II, III 4/	-	-	-	-	-	-	-	-	3.3.1	D2245	-	-
Phthalic anhydride, mass percent 4/	-	-	15.0	-	30.0	-	-	-	-	D563	-	-
Oil acids, mass percent 4/	-	-	67.0	-	47.0	-	-	-	-	D1398	-	-
Iodine number	147	175	-	-	-	-	-	-	-	D1959	-	-
Kauri reduction, percent 3/	-	-	-	-	-	-	120	150	-	D1642, Meth B	-	4.4.1.2
Rosins and rosin derivatives 3/	-	neg	-	neg	-	neg	-	neg	-	D1542, 4.1	-	-
Solids volume, percent 1/	68.0	-	59.5	-	43.0	-	50.0	-	3.5	D2697	-	-
Water, mass percent 1/	-	0.5	-	0.5	-	0.5	-	0.5	-	D4017	-	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

TABLE II. Paint. - continued.

Characteristics	Requirements I Min	I Max	II Min	II Max	III Min	III Max	IV Min	IV Max	Para	Test Methods ASTM	FTMS 141	Para
Properties												
Condition in container	-	-	-	-	-	-	-	-	3.4.1	-	3011	-
Density, kg/L (lb/gal)	2.88 (24.0)	-	2.00 (16.7)	-	2.20 (18.4)	-	2.00 (16.7)	-	3.5	D1475	-	-
Storage stability												
Skinning	-	-	-	-	-	-	-	-	3.4.2	-	3021	-
Storage stability	-	-	-	-	-	-	-	-	3.4.3	-	3022	-
Coarse particles and skins, mass percent 1/	-	1.0	-	2.0	-	1.0	-	2.0	-	-	4092	-
Fineness of dispersion, Hegman units	4	-	4	-	4	-	4	-	-	D1210	-	-
Consistency, Krebs units	75.0	89.0	73.0	86.0	72.0	88.0	74.0	86.0	3.5	D562	-	-
Viscosity, pascal seconds	-	-	-	-	-	-	-	-	3.4.4	D4287	-	-
Application												
Brushing properties	-	-	-	-	-	-	-	-	3.4.5	-	4321, 4.1	-
Spraying properties	-	-	-	-	-	-	-	-	3.4.6	-	4331	-
Drying time, hours												
Set-to-touch	-	6	-	4	1/4	1	1/4	1	-	-	4061	-
Dry-through	-	36	-	16	-	6	-	6	-	-	4061	-
Appearance and Performance of Dry Film												
Appearance	-	-	-	-	-	-	-	-	3.4.7	-	-	4.4.2.3
Flexibility	-	-	-	-	-	-	-	-	3.4.8	-	6221	4.4.2.4
Knife test	-	-	-	-	-	-	-	-	3.4.9	-	6304	4.4.2.5
Cold water resistance, type IV	-	-	-	-	-	-	-	-	3.4.10	-	2011, 2141	4.4.2.6

1/ of paint
2/ of pigment
3/ of vehicle
4/ of nonvolatile vehicle

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

3.4.1 Condition in container. The paint shall not show excessive settling in a freshly-opened full can, and shall be easily redispersed with a paddle to a smooth, homogeneous state. The paint shall show no curdling, livering, caking, or color separation, and shall be free from lumps and skins.

3.4.2 Skinning. The paint shall not skin within 48 hours in a three-quarter filled, closed container.

3.4.3 Storage stability. The paint shall show no curdling, gelling, or hard caking when stored for 12 months from the date of delivery (see 6.3.1), in a full, tightly-covered container, at a temperature of 21 to 32 degrees (°) Celsius (C) (70 to 90° Fahrenheit (F)).

3.4.4 Viscosity. The viscosity shall be the manufacturer's standard for the product provided.

3.4.5 Brushing properties. The paint, as received, shall brush easily, possess good leveling properties, and show no running or sagging tendencies when applied at a spreading rate of 500 square feet per gallon to smooth steel vertical surfaces. The dried film shall not show obvious brush marks.

3.4.6 Spraying properties. The paint shall spray satisfactorily, show no tendency to orange-peel, sag, creep, or run, and show satisfactory spraying properties in all other respects.

3.4.7 Appearance. The paint shall dry to a smooth, uniform finish, free from roughness, grit, or unevenness, and shall show no streaking or separation or surface imperfections.

3.4.8 Flexibility. The dry film of paint shall withstand bending without cracking or flaking.

3.4.9 Knife test. The paint film shall ribbon or curl from the panel when tested, and the cut shall show beveled edges.

3.4.10 Cold water resistance, type IV. When exposed to cold distilled water, the film of paint shall show no blistering or wrinkling immediately upon removal of panel. There shall be no softening of the film, and not more than a very slight whitening, when examined 2 hours after removal.

3.5 Batch verification. When specified (see 6.2), quality conformance tests (see 4.2.2.1) shall include the requirements of table III, when a first article has been previously specified. The test values, X, shall be as specified in table II, and shall be within the required ranges of the values determined in the first article testing, as specified in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

TABLE III. Batch verification.

Characteristics	Requirements
Analysis	
Pigment, mass percent	X ± 0.6
Solids volume, percent	X ± 4.0
Properties	
Density, kg/L (lb/gal)	X ± 0.05 (0.4)
Consistency, Krebs units	X ± 10.0

3.6 Material Safety Data Sheets (MSDS). MSDS shall be submitted in accordance with FED-STD-313.

3.7 Special marking.

3.7.1 Hazard marking. Each unit container shall be marked as follows:

WARNING

THIS PAINT CONTAINS LEAD AND OTHER SUBSTANCES WHICH MAY CONSTITUTE A SIGNIFICANT POTENTIAL HEALTH HAZARD TO PERSONNEL APPLYING OR REMOVING THE PAINT SYSTEM.

FOLLOW ALL PRESCRIBED HAZARD CONTROL DIRECTIVES PROMULGATED BY LOCAL HEALTH/SAFETY PERSONNEL.

COMPLY WITH SPECIFIC HEALTH HAZARD CONTROL MEASURES, PROMULGATED BY THE OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA) FOR ALL WORK PERFORMED USING THIS PAINT (29 CFR 1910.1000 and 29 CFR 1910.1025).

3.7.2 Air quality regulations marking. Each unit container and shipping container shall include the mass of VOC in grams per liter and pounds per gallon of coating, and for class 2, if not photochemically reactive, and shall also state that the material is to be used without thinning under normal environmental and application conditions.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All material must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)
b. Quality conformance inspection (see 4.2.2)
c. Preparation for delivery inspection (see 4.5).

4.2.1 First article inspection. The first article inspection shall be performed on the material when a first article is required (see 3.1 and 6.2). This inspection shall include the tests of 4.4, and the preparation for delivery inspection of 4.5. The first article may be either first production material, or standard production material from the supplier's current inventory, provided the material meets the requirements of the specification and is representative of the ingredients and manufacturing technique applicable to the remaining material to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the tests of 4.2.2.1, and the preparation for delivery inspection of 4.5, and conformance to 3.5 when specified. Additional tests may be performed as deemed necessary.

4.2.2.1 Quality conformance tests. Quality conformance inspection shall include the following test requirements of table II.

a. Analysis
   (1) Pigment mass
   (2) Red lead
   (3) Iron oxide, type II
   (4) Siliceous extender and mica, types II and IV
   (5) Solids volume

b. Properties
   (1) Condition in container
   (2) Density
   (3) Consistency
   (4) Viscosity

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

c. Application
   (1) Drying time

d. Appearance and performance of dry film
   (1) Appearance
   (2) Flexibility
   (3) Knife test
   (4) Cold water resistance, type IV

4.3 Sampling. A lot shall consist of material from a single manufacturer's batch, defined as the end product of all raw materials mixed, blended, or processed in a single operation. Inspections shall be performed on paint samples selected in accordance with ASTM D3925 (see 6.3.1).

4.4 Testing. The material shall be tested in accordance with tables I and II. Unless otherwise specified, the standard conditions and requirements of ASTM D3924 shall apply. Failure of any test shall be cause for rejection of the lot from which the sample was taken.

4.4.1 Phenolic varnish, type IV.

4.4.1.1 Preparation of varnish sample with driers and solvent. Prepare a varnish sample for designated tests by adding sufficient naphthenate driers conforming to ASTM D600 to give 0.3 percent lead, 0.03 percent cobalt, and 0.015 percent manganese, expressed as metals and based on the oils in the varnish. Sufficient solvent conforming to TT-T-291, type I, shall also be added to give a viscosity of 1.25 stokes as measured by ASTM D1545. Driers should be allowed an incorporation time of 48 hours before proceeding with any of the tests.

4.4.1.2 Kauri reduction. Substitute toluene for benzene for cleaning the test panel.

4.4.1.3 Hot water resistance. Prepare a test panel as specified in D1642 and 4.4.1.2. Flow the varnish over the panel, allowing it to spread over the entire face except for the upper 13 millimeters (mm) (0.5 inch) of one of the short edges. Allow the panel to dry in a nearly vertical position for 72 hours and then cut a 13 mm strip from the edge which was lowermost during the drying period. Then cut the panel in half, perpendicularly to the first cut, and immerse one half in a vertical position, in a beaker of boiling distilled water, for 7 hours. Remove the panel and allow it to recover for 2 hours before examination in comparison to the reserved half-panel.

4.4.1.4 Alkali resistance. Dip the closed end of a 25- by 150-mm (1- by 6-inch) test tube, cleaned with toluene, into the varnish to a depth of 100 mm (4 inches) and allow to dry, with the mouth of the tube down, for 72 hours. Place the closed end of the tube, which was uppermost during the drying period, 50 mm (2 inches) below the surface of a 5 percent sodium hydroxide

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

solution at 20 ±0.3°C (68 ±0.5°F) for 7 hours. Remove the tube and immediately rinse under gently running tap water, taking care not to touch the treated film. Dry for 2 hours and examine.

4.4.2 Paint.

4.4.2.1 Red lead. Determine red lead in accordance with FED-STD-141, method 7071 for types II and IV, and method 7073 for types I and III.

4.4.2.2 Siliceous extender and mica, types II and IV. Transfer 0.5 gram of the pigment to a 400-milliliter (mL) beaker. Add 15 mL of concentrated hydrochloric acid and heat gently until the iron oxide and red lead are completely dissolved. Evaporate to dryness and bake at 105 to 110°C (221 to 230°F), for 1 hour. Moisten the residue with a few drops of concentrated hydrochloric acid, dilute to 100 mL with hot water, boil, filter while hot, and wash with hot water. Transfer the paper and contents to a weighed porcelain crucible, ignite, cool, and weigh. Examine the residue microscopically to determine the present of mica. Calculate the percentage and report this residue as the siliceous extender and mica.

4.4.2.3 Appearance. Flow a portion of the paint on a 100- by 300-mm (4- by 12-inch) clean glass plate. Let dry in a nearly vertical position at room temperature and examine the area beginning 100 mm from the top.

4.4.2.4 Flexibility. Test as specified in FED-STD-141, method 6221, using a 3.2-mm (1/8-inch) mandrel. Draw down a wet-film thickness of 51 ±5 micrometers (0.002 ±0.0002 inch) on a 75- by 125-mm (3- by 5-inch) tinplate panel conforming to method 2012.

4.4.2.4.1 Type I. Air-dry the panel 18 hours, then bake 5 hours at 105 ±2°C (221 ±4°F).

4.4.2.4.2 Type II. Air-dry the panel 18 hours, then bake 5 hours at 75 ±2°C (167 ±4°F).

4.4.2.4.3 Type III. Air-dry the panel 3 hours, then bake 2 hours at 60 ±2°C (140 ±4°F).

4.4.2.4.4 Type IV. Air-dry the panel 18 hours, then bake 4 hours at 105 ±2°C (221 ±4°F).

4.4.2.5 Knife test. Cut the film from a flat portion of the panel used in 4.4.2.4, at a location away from the bent area, and examine.

4.4.2.6 Cold water resistance, type IV. Apply two coats of paint to both sides of a solvent cleaned panel conforming to FED-STD-141, method 2011, by brushing as specified in method 2141, allowing 24 hours air-drying between coats. Air-dry the coated panel for 96 hours, then dip the edges in a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

water-resisting varnish, and air-dry for an additional 24 hours. Immerse the panel in a beaker containing approximately 3 inches of distilled water at 21 to 32 °C (70 to 90°F), for 14 days. Remove the panel, air-dry for 2 hours, and examine.

4.5 Preparation for delivery inspection. The inspection of the packaging, packing, and marking shall be in accordance with the requirements of section 4 of PPP-P-1892.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with the requirements of PPP-P-1892, with the level of packaging and the level of packing as specified (see 6.2). The paint shall be furnished in 1-quart, 1-gallon, and 5-gallon containers as specified (see 6.2).

5.1.1 Special marking. In addition to other markings required by PPP-P-1892, containers shall be marked with special marking (see 3.7).

6. NOTES

6.1 Intended use. The four types of paint are intended for use as primers for systems in which the topcoats are to be either unmodified or modified drying oil formulations, preferably of the same generic type as the primer. The paint is not for use on residential structures. General system selection considerations are summarized in table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

TABLE IV. TT-P-86 application criteria.

Type	Application	Exposure	Substrate	SSPC Surface Preparation	Recoat Time (hr)	Typical Topcoats	Remarks
I	Shop	General atmospheric	Structural steel	SP 6	36	TT-P-102	All-linseed base wets and bonds well, even if minor corrosion remains.
	Field	General atmospheric	Structural steel	SP 2,3, or 6 1/	36	TT-P-102	All-linseed base wets and bonds well, even if minor corrosion remains.
II	Shop	General atmospheric	Structural steel	SP 6	16	TT-E-489 TT-E-490 TT-E-1593 TT-P-81 TT-P-102	Free linseed oil portion of vehicle contributes to wetting and bonding on rough steel.
	Field	General atmospheric	Structural steel	SP 3 or 6 1/	16	TT-E-489 TT-E-490 TT-E-1593 TT-P-81 TT-P-102	Free linseed oil portion of vehicle contributes to wetting and bonding on rough steel.
III	Shop/or field	General atmospheric	Structural steel/ cold-rolled steel	SP 6, minimum	6	TT-E-489 TT-E-490 TT-E-1593 TT-P-81 TT-P-102	Clean substrate required because of lesser wetting properties, with shorter drying time.
IV	Shop/or field	High humidity, and marine atmospheres	Structural steel/ cold-rolled steel	SP 6, minimum	6	TT-P-38	Clean substrate required; water resistant binder.

1/ SP 6 is better, but SP 2 and 3 may be less expensive for small areas.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Purchasers shall select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Type and class required (see 1.2)
(c) When a first article is required for inspection and approval (see 3.1, 4.2.1, and 6.3)
(d) When batch verification is required (see 3.5)
(e) Level of packaging and level of packing required (see 5.1)
(f) Size of container required (see 5.1)

6.3 First article. When a first article inspection is required, the material will be tested, and should be a sample selected from the first production material, or it may be standard production material from the contractor's current inventory, as specified in 4.2.1. The first article should consist of two gallons of material. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.3.1 Date of delivery. For the storage stability determination, the date of delivery will need to be marked on the paint sample submitted for test, or contained in the letter of transmittal.

6.3.2 Type IV paint. For type IV paint, samples of the resin, approximately one pound, and the varnish, approximately two quarts, used in the manufacture of the paint will need to be submitted in addition to a sample of the finished paint.

6.4 Shelf life and storage. Procurements for stocking should be limited to an approximate 6-month supply. Storage under normal warehouse conditions is acceptable, although long exposure to either extremely high or low temperatures should be avoided.

6.5 Typical compositions and formulas. Typical compositions and formulas for the four types of paint are described in tables V, VI, and VII.

6.6 Conformance to air quality regulations. Purchasers should specify material as described, or as otherwise required, to conform to the most stringent air quality regulations for intended jurisdictions and applications of use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

TABLE V. Pigment composition (mass percent).

Pigment	Type I		Type II		Type III		Type IV	
	Min	Max	Min	Max	Min	Max	Min	Max
Red lead (ASTM D83, 97% grade)	99.6	-	65.0	-	99.6	-	85.0	-
Pigment suspending agent	0.3	0.4	0.3	0.4	0.3	0.4	0.3	0.4
Red iron oxide pigment (min 85 percent Fe_2O_3) 1/	-	-	15	-	-	-	-	-
Magnesium silicate (ASTM D605)	-	-	-	14.7	-	-	6.0	7.0
Mica 325 mesh (ASTM D607, type A or B)	-	-	4	6	-	-	-	-
Diatomaceous silica (MIL-S-15191	-	-	-	-	-	-	7.5	8.5

1/ A siliceous type of red iron oxide

TABLE VI. Vehicle composition (mass percent) 1/.

Ingredients	Type I		Type II		Type III		Type IV	
	Min	Max	Min	Max	Min	Max	Min	Max
Raw linseed oil (ASTM D234) 2/	35	50	28	-	-	-	-	-
Heat polymerized linseed oil (TT-L-201, type II) 2/	15	30	-	-	-	-	-	-
Alkyd resin solids (TT-R-266, type III)	-	-	28	-	40	-	-	-
Phenolic varnish solids	-	-	-	-	-	-	44	-
Volatile solvent and drier	-	35	-	44	-	60	-	56

1/ Vehicle may contain appropriate amounts of antioxidants and wetting aids.

2/ Manufacturers of type I paint are cautioned to use a linseed-oil vehicle having low acid value (not over 8 based on total solids). It is important that the heat-bodied oil used in the vehicle be bodied at a low temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

TABLE VII. Paint formulas.

Ingredients (mass percent)	Type I	Type II	Type III	Type IV
Red lead (ASTM D83, 97% grade)	77.53	43.16	67.10	55.97
Red iron oxide pigment (min 85 percent Fe_2O_3) 1/	-	10.62	-	-
Magnesium silicate (ASTM D605)	-	9.76	-	4.35
Mica (ASTM D607, type A or B)	-	2.66	-	-
Diatomaceous silica (MIL-S-15191)	-	-	-	5.27
Pigment suspending agent	0.23	0.20	0.20	0.26
Raw linseed oil (ASTM D234) 2/	10.90	9.44	-	-
Pale heat-bodied linseed oil (TT-L-201, type II) 2/	3.78	-	-	-
Alkyd resin varnish (based on TT-R-266, type III)	-	18.98	27.50	-
Phenolic varnish 3/	-	-	-	25.34
Mineral spirits (TT-T-291, type I)	6.45	4.87	-	-
Aromatic petroleum spirits (TT-N-97) 4/	-	-	4.71	7.62
Dipentene (TT-D-376)	-	-	-	0.94
Antioxidants and wetting aids	-	-	0.39	0.13
Lead naphthenate (24 percent Pb) (ASTM D600, class B)	-	0.24	-	0.08
Cobalt naphthenate (6 percent Co) (ASTM D600, class B)	-	0.10	0.10	0.02
Manganese naphthenate (6 percent Mn) (ASTM D600, class E) 5/	-	-	-	0.02
Liquid drier 6/	1.11	-	-	-
Characteristics				
Pigment volume, percent	36	38	38	41
Density, kg/L (lb/gal)	2.88 (24)	2.04 (17)	2.28 (19)	2.04 (17)

1/ See table V.
2/ See table VI.
3/ A typical formula, with all percentages by weight, for varnish meeting the requirements for type IV is as follows: 100 percent p-tert-amyl phenol/formaldehyde resin 20.25, tung oil (ASTM D12) 39.75, mineral spirits (TT-T-291, type I) 40.00. These ingredients should produce a suitable varnish, when cooked according to the following schedule: Heat all the resin and oil to 240°C (465°F) in 40 minutes. Hold the batch at this temperature long enough (35 to 40 minutes) to obtain the desired viscosity when thinned to the specified solids content. Water-cool and thin with the mineral spirits. A p-tert-amyl phenol/formaldehyde resin, CK-0036, is available from Union Carbide Corp., P.O. Box 670, Bound Brook, NJ 08805.
4/ For class 2, substitution with a complying solvent system may be required.
5/ Metal concentration, 5.9 - 6.1 percent.
6/ Drying power, 1:19 raw linseed oil mix, 8 hours maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.

6.7 Subject term (key word) listing.

Air quality regulations marking
Alkyd
Batch verification
First article
Iron oxide
Linseed oil
Phenolic resin
Photochemically reactive
Primer
Red lead
Siliceous extender and mica
Solids volume
Viscometer, ICI cone/plate
Volatile organic compounds
Water resistance

MILITARY INTERESTS:

Custodian

Navy - YD

Review Activities:

Navy - AS, MS

User Activities:

Army - CE, ME

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:

Navy - YD

DOD Project No. 8010-1066

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:15Z
Check the source to verify that this is the current version before use.